

**Department of Computer Science and Engineering**  
**University of Dhaka**  
**2<sup>nd</sup> Year 2<sup>nd</sup> Semester 2025, Lab Project**  
**Course Code: CSE-2201, Database Management System**

**Part I (Design, Implementation and Report Writing)**

Consider a relational database model of your own and go for the following tasks.

1. Consider the database of reasonable number of schemas/tables (at least 8)
2. Plan for attributes that cover all general data types
3. Plan for constraints of different types (primary keys, foreign keys, unique keys, check, not null etc.)
4. Create appropriate data for the designed relations
5. Implement the database in Oracle or any DBMS system
6. Plan queries (query statement) and find the answer (at least 15 covering all the criteria mentioned in 8.g.)
7. Plan for non-trivial (canonical cover type) functional dependencies applicable for the schema design.
8. Formulate everything in a **report** which includes:
  - a. Brief description of the database system that you are implementing
  - b. Mention schemas with attributes
  - c. Show **Schema diagram** of the database
  - d. Show **E-R diagram** of the database
  - e. Snapshots of **SQL DDL** of all the schemas/tables
  - f. Snapshots of the **instances** (data of the populated tables)
  - g. **Query statement, Relational Algebra Expression, SQL statement** and snapshots of the **outputs**. SQL statements **should** contain the following :
    - i. **natural join**, cross product, **outer join**, **join with using**, **on**
    - ii. nested sub-query with clauses **some**, **all**, **any**, **exists**, **unique** etc.
    - iii. nested sub-query in **from**, **where** and **select** (scalar sub-query) clause
    - iv. **order by**, **group by**, **having** clauses
    - v. Use of **with** clause
    - vi. String manipulation, set operations
    - vii. Update, delete operations
    - viii. Use of built-in aggregate functions and other important functions
  - h. Create views and use those views in answering queries.
  - i. Find the list of non-trivial FDs and proof that the schemas are in desired normal forms
  - j. Conclusion of the work

**Part II (Web Implementation)**

Create a web application based on your designed database.

**Part III (Demonstration and Presentation)**

1. Create a presentation of the project (5 Minutes)
2. Demonstrate your project (5-8 Minutes)